



Year 11 Mathematics

Extension 1 - June 2010

Time: 60 Minutes

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.

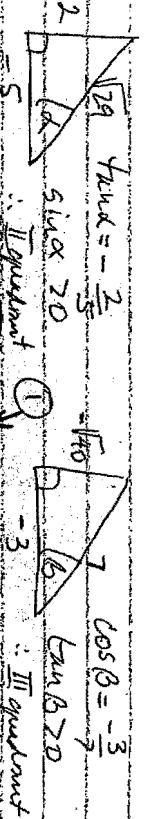
1.	Given that $\tan \alpha = -\frac{2}{5}$ when $\sin \alpha > 0$ and $\cos \beta = -\frac{3}{7}$ when $\tan \beta > 0$, find the exact value of $\tan(\alpha - \beta)$	3
2.	Solve for $0^\circ \leq x \leq 360^\circ$ a) $\cos 2x = \sin x$ b) $\sin 2x + \sqrt{3} \sin x = 0$	3 3
3.	a) Express $2 \cos x - 3 \sin x$ in the form $A \cos(x + \alpha)$ where α is acute (correct to the nearest minute). Show all working. b) Hence solve $2 \cos x - 3 \sin x = \sqrt{13}$ for $0^\circ \leq x \leq 360^\circ$	3 2
4.	Find the point P which divides the interval AB externally in the ratio 3:2, where $A(-4, -9)$ and $B(5, 9)$	2
5.	Given $A(-1, 4)$ and $B(2, -2)$. $P(x, y)$ divides AB internally in the ratio $k:1$ and P lies on $-2y - 1 = 0$. Find k .	3
6.	Given two straight lines $kx - y - 5 = 0$ and $3x + y + 1 = 0$ which intersect at 45° . Find all possible values of k .	3
7.	A committee of 5 people be selected from a group of 10 people. What is the probability that a) 2 particular members are included. b) 2 particular members are not on the committee together.	2 1
8.	a) Find how many arrangements of the letters of the word <i>PERMUTATION</i> can be formed using all letters. b) How many of these arrangements have (i) the 5 vowels together. (ii) the <i>T</i> 's are together. (iii) the consonants and vowels alternate.	1 1 1 1
9.	a) Write down the formula for $\cos 2A$ in terms of $\tan A$ b) Hence prove that if $\tan^2 y = \cos 2x$, then $\cos 2y = \tan^2 x$	1 2
10.	There are 5 boys and 4 girls to be seated at the round table. In how many ways can they be arranged so that a) Two particular girls are not together. b) no two girls are together.	1 1
11.	a) Show that the exact value of $\sin 165^\circ$ is $\frac{\sqrt{3}-1}{2\sqrt{2}}$. b) Hence, find the exact value of $\sin 82\frac{1}{2}^\circ$	2 3

~ END OF EXAM ~

Answers

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Question 1



$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{\frac{2}{5} - \frac{3}{4}}{1 + \frac{2}{5} \cdot \frac{3}{4}} = \frac{\frac{8-15}{20}}{1 + \frac{3}{10}} = \frac{-\frac{7}{20}}{\frac{13}{10}} = -\frac{7}{26}$$

Question 2

a) $\cos 2x = \sin x$
 $\cos x = \sin x$
 $1 - 2\sin^2 x - \sin x = 0$
 $(2\sin x - 1)(\sin x + 1) = 0$
 $\sin x = \frac{1}{2}$ or $\sin x = -1$
 $x = 30^\circ, 150^\circ$ or $x = 270^\circ$

b) $\sin 2x + \sqrt{3} \sin x = 0$
 $2\sin x \cos x + \sqrt{3} \sin x = 0$
 $\sin x (2\cos x + \sqrt{3}) = 0$
 $\sin x = 0$ or $\cos x = -\frac{\sqrt{3}}{2}$
 $x = 0^\circ, 180^\circ, 150^\circ, 210^\circ$

Question 3

a) $2\cos x - 3\sin x = A \cos(x + \alpha) = A \cos x \cos \alpha - A \sin x \sin \alpha$
 $2 = A \cos \alpha$
 $3 = A \sin \alpha$
 $A = \sqrt{2^2 + 3^2} = \sqrt{13}$
 $\tan \alpha = \frac{3}{2}$
 $\alpha = 56.1^\circ$

$\therefore 2\cos x - 3\sin x = \sqrt{13} \cos(x + 56.1^\circ)$

b) $\therefore \sqrt{13} \cos(x + 56.1^\circ) = \sqrt{13}$
 $\cos(x + 56.1^\circ) = 1$
 $x + 56.1^\circ = 0^\circ$ or 360°
 $x = -56.1^\circ$ or 303.9°

Question 4

$A(-4, -9)$ $B(5, 9)$
 $\therefore P\left(\frac{-4 \times 2 + 5 \times -3}{3+2}, \frac{-9 \times 2 + 9 \times -3}{-1}\right) = (2.3, 4.5)$

Question 5

$A(-1, 4)$ $B(2, -2)$
 $P\left(\frac{-1 \times 1 + 2 \times k}{k+1}, \frac{4 - 2k}{k+1}\right)$
 P lies on $x - 2y = 1$
 $\therefore \frac{-1 + 2k}{k+1} - 2 \frac{4 - 2k}{k+1} = 1$
 $-1 + 2k - 8 + 4k = k + 1$
 $3k = 10$
 $k = \frac{10}{3}$

Question 6

$kx - y - 5 = 0$ $\therefore m_1 = k$
 $3x + y + 1 = 0$ $\therefore m_2 = -3$
 $\therefore \tan 45^\circ = 1 = \left| \frac{k - (-3)}{1 + k(-3)} \right|$
 $|1 - 3k| = |k + 3|$
 $1 - 3k = k + 3$ or $1 + 3k = k + 3$
 $k = -\frac{1}{2}$ or $k = 2$

Question 7

a) $P = \frac{8C_3}{10C_5} = \frac{2}{9}$
 b) $P = 1 - \frac{2}{9} = \frac{7}{9}$
 c) $P = \frac{2 \times 4 + 8 \times 5}{10 \times 5} = \frac{7}{5}$

Question 8

$$a) \frac{11!}{2!} \text{ ①} = 19958400$$

$$b) i) \text{ (EVAID) } \dots \dots \dots 5! \frac{7!}{2!} \text{ ①} = 302400$$

$$ii) \text{ (IT) } \dots \dots \dots 10! \text{ ①} = 3628800$$

$$iii) - E - U - A - T - O - 5! \times \frac{6!}{2!} \text{ ①} = 432000 \text{ ①}$$

Question 9

$$a) \cos 2A = \frac{1 - 2\sin^2 A}{\sin^2 A + \cos^2 A} \left(\because \cos^2 A \right) = \frac{\sec^2 A - 2\tan^2 A}{\tan^2 A + 1} = \frac{1 + \tan^2 A - 2\tan^2 A}{\tan^2 A + 1}$$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A} \text{ ①}$$

$$\text{or } t\text{-formula } \tan A = t \therefore \cos 2A = \frac{1 - t^2}{1 + t^2} = \frac{1 - \tan^2 A}{1 + \tan^2 A} \text{ ①}$$

$$\text{or } \cos 2A = \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A} \left(\because \cos^2 A \right) = \frac{1 - \tan^2 A}{1 + \tan^2 A} \text{ ①}$$

$$b) \therefore \cos 2y = \frac{1 - \tan^2 y}{1 + \tan^2 y} = \frac{1 - \cos 2x}{1 + \cos 2x} \text{ ①} = \frac{1 - (\cos x - \sin x)}{1 + \cos x - \sin x}$$

$$= \frac{1 - \cos x + \sin x}{1 - \sin x + \cos x} \text{ ①} \times \frac{\sin x}{2 \cos x} = \tan^2 x \text{ ②}$$

Question 10

$$a) (G_1 G_2) \text{ are together } \therefore 2 \times 7! \text{ ①}$$

$$\therefore \text{not together } \therefore 8! - 2 \times 7! = 30240$$

$$b) 4! \times 5! \text{ ①} = 2880$$

5 boys
around
table

Girls
in
between

Question 11

$$a) \sin 165^\circ = \sin (150^\circ + 15^\circ) = \dots \text{ ①}$$

$$\text{or } = \sin (120^\circ + 45^\circ) = \sin 120^\circ \cos 45^\circ + \cos 120^\circ \sin 45^\circ$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \left(-\frac{1}{2}\right) \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{3} - 1}{2\sqrt{2}} \text{ shown}$$

$$b) \sin 165^\circ = 2 \sin 82\frac{1}{2}^\circ \cos 82\frac{1}{2}^\circ$$

$$\text{Now } \cos 165^\circ = 2 \cos^2 82\frac{1}{2}^\circ - 1$$

$$\text{and } \cos 165^\circ = -\cos (45^\circ - 30^\circ)$$

$$= -\frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= -\frac{(\sqrt{3} + 1)}{2\sqrt{2}} \quad \text{--- 1}$$

$$\therefore \cos^2 82\frac{1}{2}^\circ = \frac{1}{2} (\cos 165^\circ + 1)$$

$$= \frac{1}{2} \left(-\frac{\sqrt{3} - 1}{2\sqrt{2}} + \frac{2\sqrt{2}}{2\sqrt{2}} \right)$$

$$= \frac{2\sqrt{2} - \sqrt{3} - 1}{4\sqrt{2}}$$

$$\cos 82\frac{1}{2}^\circ = \left(\frac{2\sqrt{2} - \sqrt{3} - 1}{4\sqrt{2}} \right)^{\frac{1}{2}} \quad \text{--- 1}$$

$$\text{since } \sin 82\frac{1}{2}^\circ > 0$$

$$\text{So, } \sin 82\frac{1}{2}^\circ = \frac{\sin 165^\circ}{2 \cos 82\frac{1}{2}^\circ}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \div 2 \left(\frac{2\sqrt{2} - \sqrt{3} - 1}{4\sqrt{2}} \right)^{\frac{1}{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \left(\frac{\sqrt{2}}{2\sqrt{2} - \sqrt{3} - 1} \right)^{\frac{1}{2}} - 1$$

$$\left(\approx 0.99144 \dots \right) \text{ or equivalent.}$$